Reuben Chatterjee

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Experience

Applied Materials *Dec 2025 - Present*

Data Scientist

- Architected and developed a production grade **CV pipeline** deployed via Databricks endpoint serving, using **GLM-OCR & RT-DETR** achieving 95% precision on extraction of Part and Serial Numbers across 500+ industrial labels.
- Built a **Databricks recommender** with a **Jenkins CI/CD pipeline** and **Databricks Asset Bundles**, scoring 433K candidate operations to rank the top 5 certification routes for 2,000 technicians and writing 7,000 rows to SQL Server in under 5 minutes as a weekly scheduled job.
- Trained and evaluated an **RT-DETR** based **label detection system** by converting LabelMe annotations to COCO format, implementing custom IoU/containment matching, and achieving 97.8% precision on printed labels and 64.0% on engraved labels.
- Built a customized **LLM-powered AI agent** for a core Business Unit's sales organization, integrating Salesforce CRM, acting as an on-demand sales coach to help reps review customer sales evaluations and sharpen lead qualification.

Alcamo Marketing *Jul 2025 - Oct 2025*

Data Scientist

- **Reduced operational costs by 60% ($20K annually)** by migrating data infrastructure from Adverity to Snowflake, building Snowflake tables and views with 100+ field mappings and transformations for 4 clients.
- Enabled **real-time reporting with 99% uptime** for 4 clients by **engineering 15 automated ETL pipelines** (Fivetran to Snowflake to Looker Studio) processing 50GB+ daily data from Google Ads, LinkedIn Ads, Meta, Salesforce, and HubSpot
- Engineered **ML-based customer segmentation** with clustering algorithms **achieving 90% accuracy scores**, analyzing **behavioral patterns across 100K+ customer records** to optimize engagement strategies and prioritize high-value opportunities

Ellis Lab, UC San Diego *Sep 2024 - Jun 2025*

Graduate Research Assistant - Data Science

- Developed **automated data cleaning and validation pipeline** in R using Regex and NLP to standardize 5,000+ free-text responses, **reducing manual processing time by 90%**
- Applied **ANOVA, Tukey HSD** tests and correlation analysis to examine gender based differences male vs female dominant groups across team contributions, leadership roles, and project outcomes

Datamatics Global Services *Jun 2024 - Sep 2024*

Data Scientist Intern

- Built **Python and SQL based ETL pipelines** to process 50GB+ of economic zone data for RAKEZ, enabling demand forecasting of industrial land lease activity
- Trained a Random Forest model to optimize zone planning decisions, achieving 92% accuracy and validating results via A/B testing

Projects

Afterpay Customer Retention Prediction Model

Python, XGBoost, LightGBM, Scikit-learn, Pandas, SHAP, Snowflake, SQL

- Built **gradient boosting churn prediction model** achieving **89% accuracy and 0.92 ROC-AUC** by engineering **50+ behavioral features** including RFM metrics, payment patterns, and temporal trends from 350K+ transaction records.
- Identified **$2.1M in at-risk revenue** by **scoring 12,500 high-risk customers** through 4-tier risk segmentation system, enabling targeted retention campaigns that **improved customer lifetime value by 18%**.

Credit Card Anomaly Detection using Gradient Boosting

Python, SciKit-learn, Hugging Face, CNN, XGBoost, LightGBM, Random Forest, Seaborn

- Engineered 3,200+ behavioral features from 97,852 credit card transactions using domain-specific encodings and behavioral signals.
- Tuned LightGBM via multi-model comparison (RF, XGBoost, CatBoost), achieving 92% accuracy and 0.59 OOT AUC
- Reduced false positives by 10% via threshold tuning and SMOTE, contributing to $2M+ projected annual savings

Education

University of California, San Diego

Master of Science in Data Science (CGPA:3.82) *Sep 2023 - Jun 2025*

Courses: Statistical Models, Scalable Data Systems, Fraud & Pricing Analytics, Deep Learning

University of Mumbai

Bachelor of Engineering in Computer Engineering (CGPA:3.93) *Aug 2019 - May 2023*

Courses: Data Structures, Big Data Analytics, DBMS, AI & Machine Learning, NLP

Technical Skills

Programming: Python, SQL, R, C++, JavaScript

Machine Learning & AI: PyTorch, Hugging Face Transformers, Langchain & LangGraph, LLM, VLM, OCR, Object Detection (RT-DETR), Model Evaluation (Precision/Recall/F1, IoU)

Data Engineering & MLOps: ETL & Batch Pipelines, CI/CD (Jenkins, Databricks Asset Bundles), Model Deployment & Inference (Databricks, AWS, Snowflake), GPU Optimization (float16, NVMe caching), Annotation Formatting (COCO, LabelMe), Batch Scheduling, Workflow Orchestration, CLI, Git

Data Platforms & Tools: Snowflake, Databricks, JIRA, AWS, Azure, Hadoop, dbt, Docker, Fivetran

Visualization & Communication: Tableau, Looker Studio, Power BI, Excel, ggplot2, Shiny, Seaborn